

Task 2: California House Price Prediction – Model Comparison

Submitted by: Adyasha Sahu

Internship: AI & Machine Learning Internship

Organization: Main Crafts Technology

1. Objective

The objective of this project was to compare multiple machine learning regression models for predicting California house prices using the California Housing Dataset. The project involved preprocessing the dataset, applying feature scaling, training different regression models, evaluating their performance using standard metrics, selecting the best-performing model, and deploying the final model through a Streamlit application.

2. Dataset

The California Housing Dataset, available in the Scikit-learn library, was used for this project. The dataset contains housing-related information collected from different districts of California.

The target variable is **HousePrice (Median House Value)**, while the input features include:

- Median Income, House Age, Average Rooms, Average Bedrooms, Population, Occupancy, Latitude, Longitude,
- The dataset was checked for missing values and data types before model training.

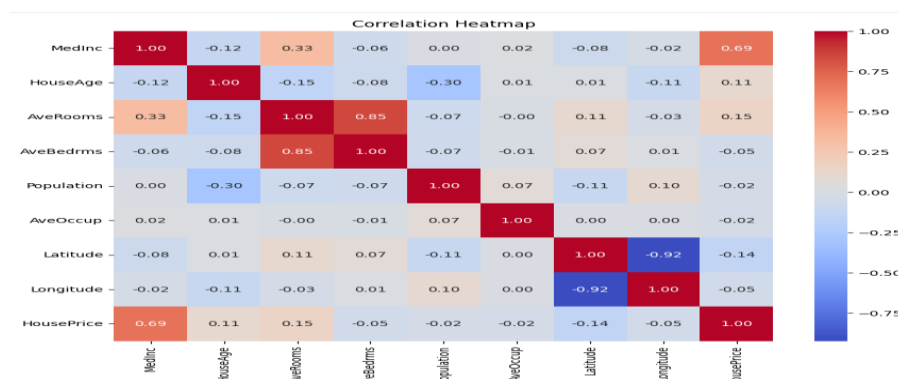
3. Technologies Used

- Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib, Streamlit

4. Methodology

The project followed a complete machine learning workflow consisting of the following steps:

1. Imported the required Python libraries.
2. Loaded the California Housing Dataset.
3. Performed Exploratory Data Analysis (EDA).
4. Visualized feature distributions and correlations.
5. Applied feature scaling using StandardScaler.
6. Split the dataset into training and testing sets.
7. Trained multiple regression models.
8. Compared model performance using evaluation metrics.
9. Selected the best-performing model.
10. Saved the trained model and scaler using Joblib.
11. Developed a Streamlit application for predicting house prices.



5. Machine Learning Models Compared

The following regression models were trained and evaluated:

Model	Purpose
Linear Regression	Baseline regression model
Ridge Regression	Linear model with L2 Regularization to reduce overfitting
Decision Tree Regressor	Captures complex and non-linear relationships in the dataset

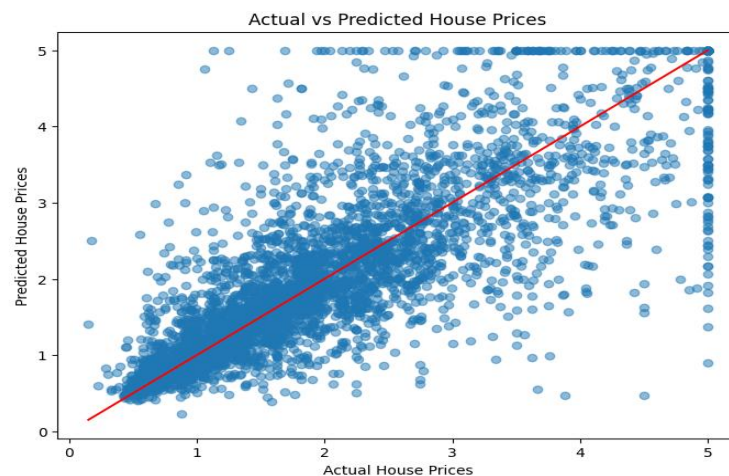
6. Model Performance

The models were evaluated using three standard regression metrics:

- **MAE (Mean Absolute Error):** Measures the average prediction error.
- **RMSE (Root Mean Squared Error):** Gives higher importance to larger errors.
- **R² Score:** Measures how well the model explains the variance in the target variable.

Model	MAE	RMSE	R ² Score
Linear Regression	0.5332	0.7456	0.5758
Ridge Regression	0.5332	0.7456	0.5758
Decision Tree Regressor	0.4531	0.7022	0.6237

The Decision Tree Regressor achieved the best overall performance among the evaluated models.



7. Best Model Selection

After comparing all three models, the **Decision Tree Regressor** was selected as the best-performing model.

The final model achieved:

- **MAE:** 0.4531
- **RMSE:** 0.7022
- **R² Score:** 0.6237

The trained model was saved as **best_house_price_model.pkl**, while the fitted StandardScaler was saved as **scaler.pkl**. These files were later used in the Streamlit application for making predictions on new user inputs.

```
=====
🏆 Best Performing Model
=====
Model      : Decision Tree
MAE        : 0.4531
RMSE       : 0.7022
R2 Score  : 0.6237
```

8. Results

The comparison of multiple regression models demonstrated that different algorithms perform differently on the same dataset. Among the evaluated models, the Decision Tree Regressor achieved the lowest prediction error and the highest R² Score.

The project also demonstrated the importance of data preprocessing, feature scaling, model evaluation, and model selection in developing effective machine learning solutions.

The Streamlit application further enhanced the project by providing an interactive interface for predicting house prices using the trained model.

9. Future Improvements

The project can be enhanced further by:

- Performing hyperparameter tuning.
- Applying Cross Validation for more reliable evaluation.
- Comparing additional models such as Random Forest and XGBoost.
- Performing Feature Engineering.
- Deploying the Streamlit application on Streamlit Community Cloud.

10. Conclusion

This project provided practical experience in building, evaluating, and comparing multiple machine learning regression models. It covered the complete machine learning workflow, including data preprocessing, feature scaling, model training, evaluation, model comparison, model persistence, and deployment using Streamlit.

The Decision Tree Regressor emerged as the best-performing model for the California Housing Dataset. This project strengthened my understanding of regression algorithms, evaluation metrics, and real-world machine learning development practices.